

Wayfarer's Crawl: An Enemy AI Focused Game

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Project Goal

Wayfarer's Crawl focuses on the decision-making logic of enemies. This is done in the Godot engine by using:

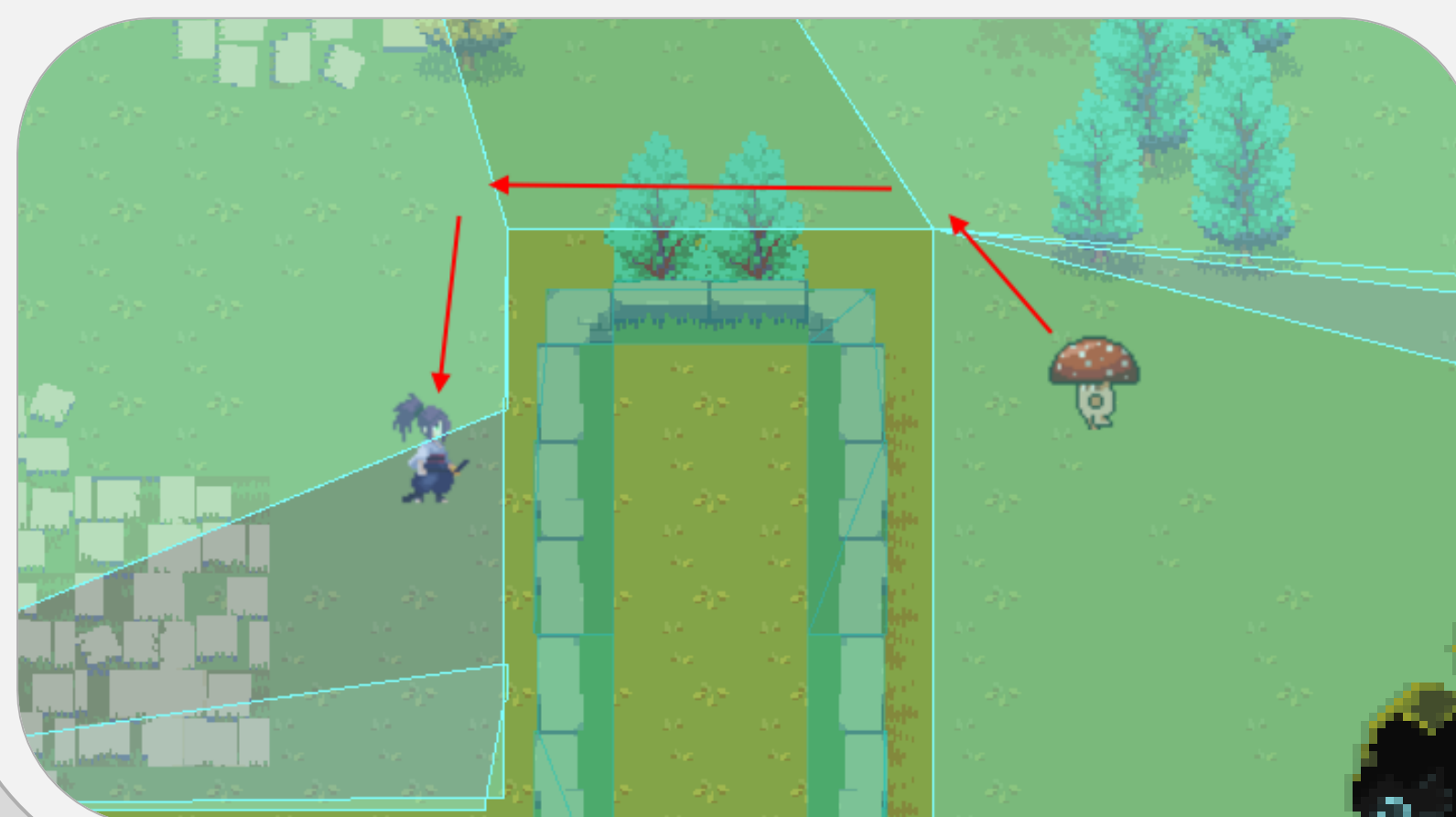
- Pathfinding
- State Machines
- Behavior Trees



Pathfinding

Enemies use a Navigation Mesh to find the optimal path to the player. This ensures enemies stay within walkable areas and navigate around obstacles. A Navigation Mesh is used to implement these features:

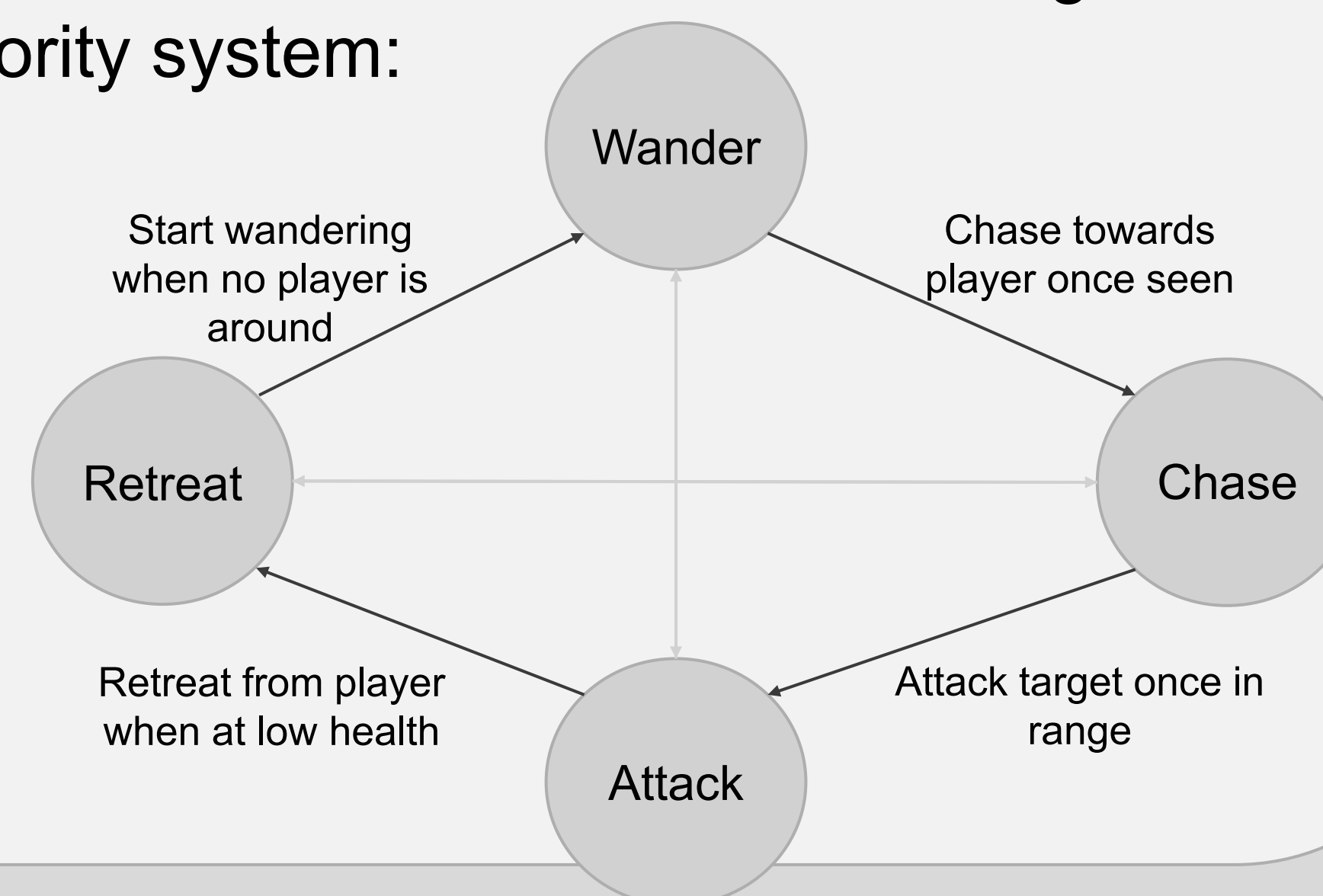
- A* algorithm
- Recalculate paths in real-time
- Scaling to support multiple enemies
- Provide smooth enemy navigation



Finite State Machine

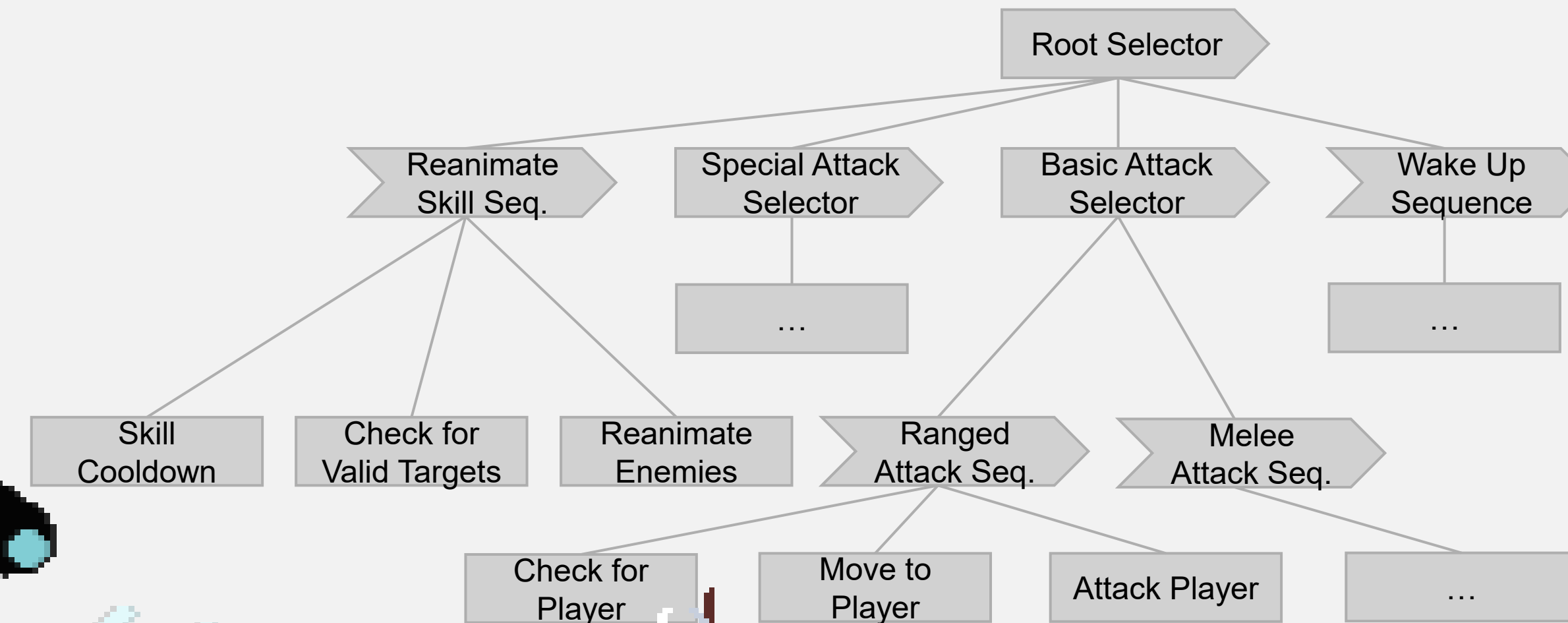
States machines are a tool used by the enemies to control their actions. Each enemy exists in one state at a time, where each state has its' own logic. Transitions between states occur based on conditions being met and have a priority system:

- Hurt
- Retreat
- Attack
- Chase
- Wander
- Idle



Behavior Tree

Another tool used to control enemy logic are behavior trees, which focus on organizing actions into a hierarchical structure of nodes. These differ from state machines by offering a controlled sequence of actions.



Player Classes

Two player classes, one for melee combat and another for ranged:

- Samurai – attacks that do bleed damage to enemies
- Archer – skills based around stunning the enemies



State Visualization

A visualization tool used to inspect and control enemy behavior:

- Text-based and graphical view
- Used to test states in game

